

DATA CLEANING ON LAPTOP DATA

1. Backup Created

A copy of the original dataset was created to keep the original data safe and avoid losing data by accident.

2. Checked Rows and Memory

The number of rows in the table and how much memory it uses were checked to understand the size of the data. It calculates the amount of memory space (in bytes and kilobytes) used by dataset in the database.

- Before Cleaning: Memory usage: 262144 bytes (approx. 263 KB).
- After Cleaning: Memory usage: 196608 bytes (approx. 196 KB).

It is observed that the cleaning process reduced memory usage by 65536 bytes (64 KB). The drop in memory consumption is likely due to:

- Dropping unnecessary columns.
- Removing duplicate rows.
- Changing data types to more efficient ones.
- Eliminating null values and unused data.

3. Removed Unnecessary Data

- The column Unnamed: 0 was deleted because it was not useful.
- Checked if any values were missing (none were found).
- Duplicate rows with the same details were removed.

4. Fixed Data Types

- Changed screen size (Inches) to a decimal number with one decimal place.
- Removed "GB" from RAM values and made them whole numbers.
- Removed "kg" from Weight values and fixed missing weights (marked as "?").
- Rounded Price values to whole numbers.

5. Simplified Operating System (OpSys)

Operating systems were grouped for simplicity such as:

- Mac systems → "macos"
- Windows versions → "windows"
- Linux versions → "linux"
- No OS → "NA"
- Android & Chrome OS → "other"

6. Split GPU

- The gpu_brand column was created from Gpu
- It stores brand names of graphics cards such as NVIDIA, AMD or Intel.

7. Split CPU

- The Cpu column was split into cpu_brand, cpu_name, and cpu_speed.
- This helps to analyse processor characteristics independently.

8. Screen Resolution Cleaned

- Screen resolution was split into ResolutionWidth and ResolutionHeight.
- A new column Touchscreen was added to show if the laptop has a touchscreen and later removed the original ScreenResolution column.

10. Memory Cleaned

- Divided Memory column into MemoryType, PrimaryStorage, and SecondaryStorage.
- MemoryType includes SSD, HDD, HYBRID (both SSD and HDD), or FLASH STORAGE.
- PrimaryStorage stores the main storage size, while SecondaryStorage is set if a + indicates additional storage; otherwise, it is 0.

- Converted all storage sizes to GB by multiplying TB values by 1024.
- Removed laptops with missing storage data.
- Deleted the original Memory column after extraction.

11. Final Cleanup

- Unnecessary columns such as Gpu, Cpu, and Memory were dropped after extracting useful information. Null values and duplicates were removed.